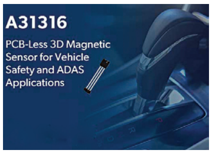
matrix computations. Memory bandwidth is optimised by a single unified compilation stack that helps result in significant power minimisation. Chimera can run any ML model easily.
Quadric
https://quadric.io

PCB-less 3D Hall Effect position sensor

Allegro A31316 is a PCB-less 3D Hall Effect position sensor that comes in a small 4mm × 4 mm PCB-less package. The IC includes the industry's fastest SENT protocol with 0.5µs to 10µs tick times and includes a pulse-width modulation (PWM) interface with 125Hz to 16kHz carrier for simple digital interface to microcontrollers. The A31316 is ideal for automotive safety and advanced driver assistance systems (ADAS) that require high levels of flexibility and reliable performance in harsh conditions with extreme vibrations such as powertrain of vehicle, and demanding industrial applications that require linear or rotary sensing. This sensor integrates the die and capacitors in a single package to address the need for high-performance, compact position sensors that are robust. It can enable flexible integration for precision contactless position sensing



Allegro MicroSystems, Inc.
https://www.allegromicro.com

Smallest behind-the-year implant sound processor

The nucleus 8 sound processor is the world's smallest and lightest behind-the-ear implant sound processor. It can sense changes in the environment and automatically adjust the settings for more immersive sound, thus improving the sound quality while enhancing the comfort. It helps users hear conversations more clearly and easily, particularly in noisy situations. The nucleus 8 also supports the next-gen Bluetooth technology which enables you to connect directly to broadcast at public venues such as airports, conference centres, and theatres using the latest Auracast technology. It also enables firefighters and other rescuers to clearly communicate with the survivors and colleagues in noisy environments.

Cochlear
https://www.cochlear.com

Compact SATCOM module for integrated IoT systems

Astronode S from Astrocast is a compact satellite communication (SATCOM) module for highly integrated IoT systems. It is a low-power module that enables bi-directional satellite communication on a single battery for a period of up to 10 years. It features a UART serial interface and 2-level 256-bit AES encryption with unique device keys. This module is available in a surface-mount package that measures 35mm × 31mm for trouble-free integration and soldering onto a PCB. It is ideal for use in a wide field of applications, including agriculture (fuel management, precision farming), maritime applications (container tracking, fishing buoys), and environment (weather data, flow monitoring).

Astrocast
https://www.astrocast.com

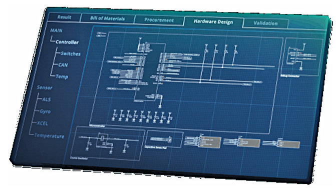
SPoE solutions for greater intelligence in smart buildings

LTC4296-1 and LTC9111 from Analog Devices are single-pair power over Ethernet (SPoE) solutions that enable greater levels of intelligence in smart buildings. Both LTC4296-1 and LTC9111 can reliably transfer power and data over one kilometer, which is a significant increase from previous Ethernet standards. The SPoE solution augments single-pair Ethernet (SPE) to provide more reliable, fault-tolerant, and interoperable point-to-point power solutions. It delivers power up to 52W. Both products are 802.3cg compliant and support Serial Communication Classification Protocol. The devices can help endpoint applications to be seamlessly powered and accessed across the network and used to assess local factors like asset health, environmental conditions, and security metrics. They reduce reliance on localised power and batteries by using a single twisted pair of Ethernet wires to provide efficient, reliable, easily installed power at reduced size and weight.

Analog Devices Inc.
https://www.analog.com

AI based error-free electronics design tool

ACE by Circuitmind is an AI based tool that can enable electronics design engineers to create and verify high-quality schematics and generate PCB layouts. The smart algorithm can select suitable converter topologies and automatically choose optimised components based on the requirements. The AI tool allows generation of multiple design options, trading off amongst size, cost, and power. ACE uses the latest procurement information to indicate sources for the required components, thus reducing the time to market. It mathematically generates error-free designs and executes a series of independent rule checks on the circuits to guarantee errorless designs. The smart AI tool can enable even inexperienced engineers to design high-level schematics in a short time and reduce BoM count, cost, and the time required in prototyping phase.



Circuitmind
https://www.circuitmind.io